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Kévin Verdière; Raymond Panneton; Saïd Elkoun; Thomas Dupont; Philippe Leclaire



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Kévin Verdière,^{a)} Raymond Panneton, and Saïd Elkoun

GAUS, Department of Mechanical Engineering, Université de Sherbrooke (QC),
J1K 2R1, Canada

*Kevin.Verdier@USherbrooke.ca, Raymond.Panneton@USherbrooke.ca,
Saïd.Elkoun@USherbrooke.ca*

Thomas Dupont and Philippe Leclaire

DRIVE, ISAT, Université de Bourgogne, 49 Rue Mademoiselle Bourgeois,
58027 Nevers Cedex, France

Thomas.Dupont@u-bourgogne.fr, Philippe.Leclaire_isat@u-bourgogne.fr

Abstract: A transfer matrix method to predict absorption coefficient and transmission loss of parallel assemblies of materials which can be expressed by a 2×2 transfer matrix was published recently. However, the usual method based on the sum of admittances is largely used to predict also surface admittance of parallel assemblies. This paper aims to highlight differences between both methods through three examples on a parallel assembly backed by (1) a rigid wall, (2) an air cavity, and (3) an anechoic termination.

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1. Introduction

A parallel assembly is a stack of acoustic systems parallel to the incident wave propagation direction. It differs from the serial case where the acoustic systems are stacked one after another along the incident wave propagation direction. Admittance sum method (ASM)¹ is largely used to evaluate surface admittance of a simple parallel construction of different acoustic systems such as a parallel microperforated panel.² Complex constructions such as inclusion in porous media³ have to be modeled by the finite element method (FEM) or an analytical solution. A new transfer matrix method has been recently presented⁴ for assembling the elements of a parallel construction into a global 2×2 transfer matrix, which may in turn be assembled in series with another 2×2 transfer matrix of a serial or parallel construction. It can be a simpler way to model these complex structures. This Express Letter aims at highlighting the differences between the parallel transfer matrix method (P-TMM) and the ASM. In this paper, a parallel assembly was studied with three different backing conditions: Rigid wall, air cavity, and anechoic termination. The results are presented in terms of the normal incidence sound absorption coefficient, and compared with the FEM.

2. Theory

Assume a parallel assembly of n porous cells which can be expressed by a 2×2 transfer matrix T_i . In the following, the ASM and the P-TMM are presented. The assumptions for both methods are the same: Each cell is locally reacting and the patchwork so formed is much smaller than the acoustic wavelength to consider it as homogeneous.

^{a)} Author to whom correspondence should be addressed.

Table 1. Surface admittance of cell i according to the backing condition. Z_0 and k_0 are the characteristic impedance and wave number in air, respectively. $T_{i, nm}$ is the transfer matrix coefficient (line n and column m) of cell i and d is the thickness of the air cavity.

Backing condition	Close-ended or rigid wall	Anechoic	Air cavity of depth d
Admittance Y_i	$Y_i = \frac{T_{i,21}}{T_{i,11}}$	$Y_i = \frac{T_{i,21} + \frac{T_{i,22}}{Z_0}}{T_{i,11} + \frac{T_{i,12}}{Z_0}}$	$Y_i = \frac{T_{i,22} + jZ_0 \cot(k_0 d) T_{i,21}}{T_{i,12} + jZ_0 \cot(k_0 d) T_{i,11}}$

2.1 ASM

The ASM is a classical method to obtain surface admittance of a parallel assembly made of several locally reacting elements.¹ The global surface admittance is the sum of the surface admittances of the cells forming the patchwork weighted by their surface ratios, respectively. It is given by

$$Y_G = \sum_{i=1}^n r_i Y_i, \quad (1)$$

where r_i is the surface ratio between cell i and the patchwork, and Y_i is the surface admittance of cell i given by Table 1 according to the backing condition. In Table 1, Z_0 and k_0 are the characteristic impedance and wave number in air, respectively.

Knowing the global surface admittance of the patchwork, the sound absorption coefficient can be deduced by

$$\alpha = 1 - \left| \frac{1 - Y_G Z_0}{1 + Y_G Z_0} \right|^2. \quad (2)$$

2.2 P-TMM

P-TMM is a recent method to evaluate absorption coefficient and transmission loss of a parallel assembly expressed as a 2×2 transfer matrix. The detail of the method is given in Ref. 4. In this work, two cases are studied: Assembly of close-ended cells and assembly of open-ended cells. Equations (3) and (4) give the global transfer matrix $\mathbf{T}_p$ of each case, respectively,

$$\mathbf{T}_p = \frac{-1}{\sum r_j y_{j,21}} \begin{bmatrix} \sum r_j y_{j,22} & -1 \\ \sum r_j y_{j,12} \left(\sum r_i y_{i,11} - \sum r_k \frac{y_{k,12} y_{k,21}}{y_{k,22}} \right) - \sum r_j y_{j,12} \sum r_j y_{j,21} & \sum r_k \frac{y_{k,12} y_{k,21}}{y_{k,22}} - \sum r_i y_{i,11} \end{bmatrix} \quad (3)$$

$$\mathbf{T}_p = \frac{-1}{\sum r_j y_{j,21}} \begin{pmatrix} \sum r_i y_{i,22} & -1 \\ \sum r_i y_{i,22} \sum r_i y_{i,11} - \sum r_i y_{i,12} \sum r_i y_{i,21} & -\sum r_i y_{i,11} \end{pmatrix}. \quad (4)$$

In Eqs. (3) and (4), subscripts i , j , and k , respectively, correspond to all the cells, open-ended cells only, and close-ended cells only, and the $y_{i, nm}$ terms are the admittance matrix coefficient of cell i . These coefficients are related to the 2×2 transfer matrix coefficients $t_{i, nm}$ of cell i by

$$\mathbf{Y}_i = \begin{bmatrix} y_{i,11} & y_{i,12} \\ y_{i,21} & y_{i,22} \end{bmatrix} = \frac{1}{t_{i,12}} \begin{bmatrix} t_{i,22} & t_{i,21} t_{i,12} - t_{i,22} t_{i,11} \\ 1 & -t_{i,11} \end{bmatrix}. \quad (5)$$

For instance, the 2×2 transfer matrices for an air layer and a rigid porous medium of thickness h are, respectively,⁵

$$T_{\text{air}} = \begin{bmatrix} \cos(k_0 h) & jZ_0 \sin(k_0 h) \\ \frac{j \sin(k_0 h)}{Z_0} & \cos(k_0 h) \end{bmatrix}, \quad (6)$$

$$T_{\text{rigid porous}} = \begin{bmatrix} \cos(k_{\text{eq}} h) & jZ_{\text{eq}} \sin(k_{\text{eq}} h) \\ \frac{j \sin(k_{\text{eq}} h)}{Z_{\text{eq}}} & \cos(k_{\text{eq}} h) \end{bmatrix}, \quad (7)$$

where Z_{eq} and k_{eq} are the characteristic impedance and wave number in the porous medium. Once the global transfer matrix of the patchwork is known, its sound absorption coefficient is given by

$$\alpha = 1 - \left| \frac{\frac{T_{p,11}}{T_{p,21}} - Z_0}{\frac{T_{p,11}}{T_{p,21}} + Z_0} \right|^2. \quad (8)$$

3. P-TMM versus ASM

In order to compare both methods, a patch of three cells filled with porous materials is modeled by FEM using Comsol software. The FEM model consists of a numerical representation of the standard two-microphone impedance tube.⁶ Each cell represents a third of the construction and the patch is 2 cm thick. Porous materials are modeled as equivalent fluids using the Johnson-Champoux-Allard model.⁵ It requires five parameters which are porosity, air flow resistivity, tortuosity, viscous, and thermal characteristic lengths. Their properties are given in Table 2. Both methods assume lateral independence of cells. Consequently, in the FEM model, interior impervious wall conditions are inserted between each cell. Three cases are studied with different backing conditions: Rigid termination, air cavity, and anechoic termination. As the ASM only gives surface impedance, the results are expressed in terms of the sound absorption coefficient. The transmission loss can only be calculated by P-TMM.

3.1 Parallel assembly on rigid termination

The study starts with the patchwork on a rigid termination. Two cases are presented in Fig. 1. The first case is the ideal assembly where the rigid backing is perfectly sealed on the patchwork. The second case corresponds to the patchwork unsealed on the rigid backing. This second case is often more representative of the real measurement, insofar as acoustic leaks can exist. For example, they can be due to an uneven surface material or to the cutting process. The unsealed case is modeled by adding a tiny air layer of 0.5 mm between the patchwork and the rigid termination. Figures 2(a) and 2(b) present the sound absorption coefficients as a function of frequency predicted by ASM,

Table 2. Physical properties of selected materials according to the Johnson-Champoux-Allard model.

	Material	Porosity	Air flow resistivity (N s m ⁻⁴)	Viscous length (μm)	Thermal length (μm)	Tortuosity
A	Melamine foam	0.999	10 000	110	122	1.00
B	Glass wool	0.999	15 957	97	530	1.00
C	Polyurethane foam	0.958	11 188	70	209	1.94

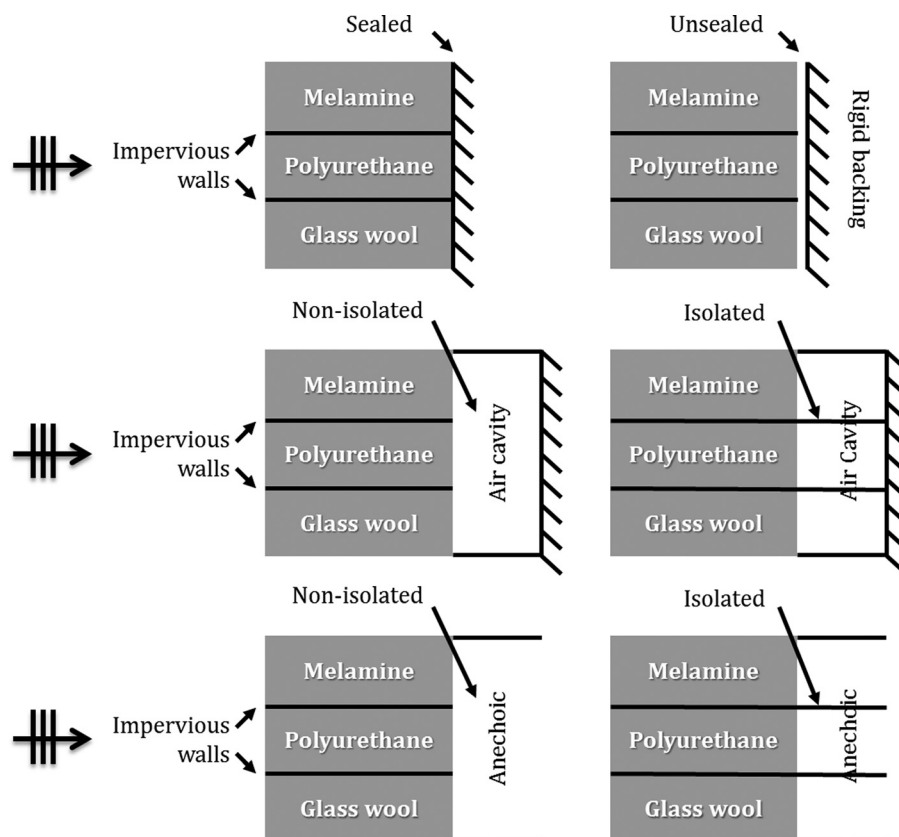


Fig. 1. Sketch of the parallel construction backed on a rigid wall (top), on a 1 cm thick air cavity (middle), and an anechoic termination (bottom). The rigid termination can be sealed or unsealed to the construction. The air cavity and the anechoic termination can be divided into three independent isolated extensions of each cell.

P-TMM, and FEM for both cases. For the sealed case, the three methods give the same results. Here, the close-ended P-TMM version has been used. For the unsealed case, a discrepancy between ASM and FEM can be noticed. In fact, ASM sees only the front of the construction whereas P-TMM takes into account the back. Moreover, ASM assumes that each cell is always independent: It is not possible to have leaks. In this case, the open-ended P-TMM version was used and combined in series with the air cavity transfer matrix to account for the tiny air layer: A pressure communication exists. That is why the unsealed FEM and PTMM results match perfectly. As shown, a slight acoustical leak in the parallel construction can change significantly the acoustic response (first conclusion). Moreover, a combination of serial and parallel TMM can give the same results of ASM in Fig. 2(b). Each porous material is combined with the tiny air cavity to form a serial transfer matrix (S-TM), then the global parallel transfer matrix is formed by each S-TM using close-ended P-TMM. In this way, P-TMM includes ASM and can model a more complex system (second conclusion).

3.2 Parallel assembly with air cavity backing condition

In this section, the parallel construction is backed by a 1 cm thick air cavity (Fig. 1). Figure 2(c) presents the sound absorption coefficient given by ASM, P-TMM, and FEM. Both ASM and P-TMM have the same independence rule between cells; however, P-TMM can be combined in series with another transfer matrix and ASM only sees the front of the construction. Consequently, to compare FEM with ASM, the air cavity has to be divided into three independent parts in the FEM model by adding

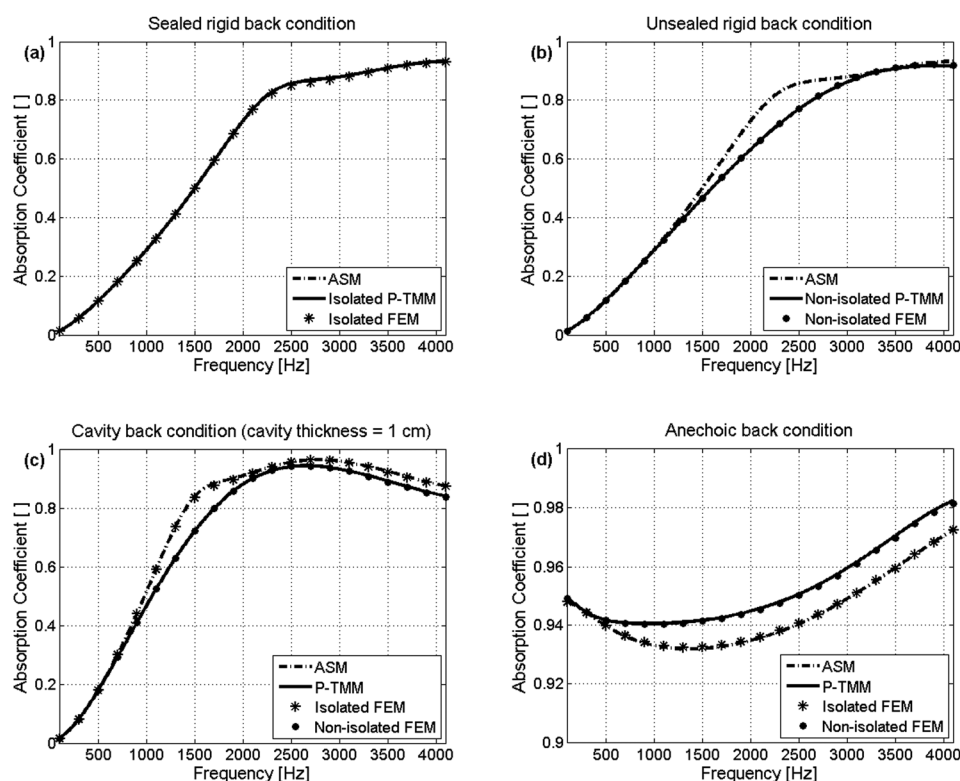


Fig. 2. Sound absorption coefficients predicted by ASM, P-TMM, and FEM: For the sealed (a) and unsealed (b) cases; for the isolated and non-isolated air cavity cases (c); for the isolated and non-isolated anechoic cases (d).

impervious walls in the extension of each cell (Fig. 1). This case corresponds to the isolated FEM curve in Fig. 2(c). The non-isolated FEM curve corresponds to the global air cavity backing the construction (Fig. 1). This FEM solution and the P-TMM results are in agreement. Similarly, the ASM and isolated FEM results fit perfectly.

To conclude, the independence rule between cells has to be respected only in the patchwork for P-TMM and in the entire assembly (patchwork and backing cavity) for ASM.

3.3 Parallel assembly with anechoic termination

In this section, the construction is backed on an anechoic termination (Fig. 1). Figure 2(d) presents the sound absorption coefficient given by ASM, P-TMM, and FEM. As the back condition is anechoic, the ordinate scale is focused between 0.9 and 1 in order to see the discrepancies between ASM and P-TMM. As in Sec. 3.2, two FEM solutions have been done with and without impervious walls in the anechoic part. For the same reasons as in Sec. 3.2, PTMM and non-isolated FEM results are in agreement, whereas ASM and isolated FEM results are in line. ASM assumes the independence of cells all along the entire assembly.

4. Conclusion

This paper presented the differences between P-TMM and ASM in terms of the sound absorption coefficient. To validate each approach, the FEM was used. It was shown that if no acoustical leaks exist between the patchwork and the rigid termination, ASM and P-TMM give the same results. However, if leaks exist between the parallel assembly and the rigid termination, P-TMM is able to model correctly the acoustic

response by adding a tiny air layer between the patchwork and the rigid termination. Particular attention should therefore be paid to the best modeling choice of the rigid termination with or without acoustical leaks.

It was also shown that if the construction is backed by an air cavity or an anechoic termination, P-TMM is the only method (compared with ASM) that gives the correct acoustic response. This is due to the fact that ASM always assumes that the backing is divided into separate cells which are an extension of the cells of the patchwork.

In conclusion, P-TMM is a method with a better potential than ASM for predicting acoustic surface properties because P-TMM can model with a lot of versatility combinations of objects of different sizes and placed in parallel and/or in series. Moreover, with a specific arrangement, P-TMM includes ASM and can also be used to predict sound transmission loss. Future works are in progress such as validating a reverse method to predict acoustic properties of each cell from measurements of its parallel construction.⁷

Acknowledgments

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